SARVESH S

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PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering student (2027) with a strong foundation in VLSI design, FPGA-based computing, Embedded Systems, and AI/ML hardware acceleration. Experienced in developing energy-efficient CNN accelerators through dynamic sparsity optimization during a research internship at NIT Tiruchirappalli. Proficient in Verilog, Xilinx Vivado, Python, C, Arduino, and ESP32, with hands-on experience in RTL design, hardware prototyping, and IoT-based system development. Passionate about digital hardware innovation, FPGA implementation, and designing efficient, scalable engineering solutions.

TECHNICL SKILLS

Programming Languages: Embedded C, C++, Python, Verilog

Hardware & Tools: Xilinx Vivado, STM32 Microcontrollers, Rasperry Pi, Kriaa KV260, Arduino

Domains & Concepts: VLSI Design, Embedded Systems,, Digital Electroncs, CNN, DSC, AI&ML Hardware Interface

Frameworks/Software: MATLAB & Simulink, PSpice & LTspice, Vitis Hls,Arduino IDE, VS code, PyTorch

EDUCATION

Saranathan College of Engineering, Trichy 2023 - 2027

Bachelor of Engineering in Electronics and Communication Engineerring

WORK EXPERIENCE

Hardware Intern Jun 2026 – Jul 2026

National Institute of Technology Tiruchirappalli

- Developed an adaptive CNN accelerator architecture using dynamic precision and sparsity optimization for energy-efficient FPGA inference.
- Implemented sparsity detection and zero-skipping techniques to reduce redundant computations and MAC operations.
- Explored Depthwise Separable Convolution (DSC) and approximate computing techniques to improve computational efficiency.
- Developed and evaluated CNN models using Python and PyTorch, with MNIST dataset-based testing..

PROJECTS

Data Insights and Strategic Unit (DISU) – IoT-Based Postal Monitoring System

- Developed an IoT-based postal monitoring prototype for real-time parcel tracking and divisional performance monitoring.
- Integrated ESP32, Raspberry Pi, and multiple sensors to monitor parcel entry/exit, weight, temperature, and security movements..
- Implemented sensor data acquisition and real-time data transmission using embedded systems and backend APIs..
- Integrated MongoDB for data storage and analytics to support centralized monitoring and operational insights.

Leadership & Positions of Responsibility

President - IEI Student Chapter

Saranathan College of Engineering, Trichy

- Leading IEI student chapter activities and coordinating technical initiatives and student engagements..
- Organized and managed SPACE SPECTROM, a project expo, encouraging innovation and technical knowledge sharing among students.
- Coordinated student teams, faculty members, and event activities to ensure successful execution of technical events.